AMD-Optimized-Qwen-VL-PartialInpainting

AMD ROCm Optimized Partial Inpainting

INFERENCE SPEED BOOST

80s → 25s

⚡ 3x up↑

VRAM CONSUMPTION

↓ 40% Reduction

Optimized for AMD GPUs

- One-Click Deploy
- Fully Local
- Open Source & Free

Presented by : gaojie | Track 1 : Multi-modal Content Creation Tool Development

Pain Points

1. Prohibitive Cloud Costs

• Pay-per-use models drain budgets for heavy users
• Bulk processing leads to inflated monthly fees
• Long-term reliance creates unsustainable overhead

2. Privacy Risks

• Raw assets exposed to third-party cloud servers
• Corporate IP, designs & portraits at risk
• Non-compliance with GDPR/CCPA regulations

3. AMD Hardware

• Severe VRAM fragmentation on ROCm
• Slow native inference (80s/image)
• Frequent OOM crashes break workflows

AMD GPU users deserve a**high-performance, free, and privacy-first**local image editing tool—empowering creativity without compromising security or breaking the bank.

What did we do?

BEFORE · Native & Unoptimized

80s Inference

Slow response speed with high latency, unable to handle real-time demands.

50GB VRAM

Excessive memory footprint limits the number of concurrent tasks.

Frequent OOM

Unstable operation and random crashes disrupt service continuity.

Manual Setup

Complex environment configuration and high technical barriers.

AFTER · Our Optimized Solution

25s (3× Faster)

Significant reduction in inference time, enabling smooth real-time interaction and high throughput.

30GB (↓40%)

Optimized memory management reduces usage by 40%, allowing more concurrent instances per GPU.

100% Stable

Memory fragmentation reduced by 90%, eliminating OOM errors for 7x24h reliable service.

1-Click

Zero-config, one-click installation script gets you up and running in minutes.

Four commonly used creative templates

1. Style Redrawing

Transform parts or the entire image into diverse artistic styles—from cyberpunk futurism to classic oil painting textures—realizing creative style migration with high visual fidelity.

2. Garment Replacement

Instantly swap apparel on models while preserving natural body poses and lighting. It supports reference image guidance for precise fabric texture and style matching, optimizing fashion design workflows.

3. Text-Guided Generation

Generate or modify specific image areas using natural language prompts. Define details like objects, colors, and backgrounds with simple text, achieving intuitive and precise content customization.

4. Object Removal

Effortlessly erase unwanted elements like watermarks, passersby, or distracting clutter. The AI intelligently fills the void with contextually coherent details for a seamless, professional finish.

System Architecture

1 USER INTERACTION LAYER

Provides an intuitive visual interface for model tuning, mask editing, and real-time result preview.

- **Gradio UI** — Visual Web Portal
- **Mask Edit** — Interactive Drawing
- **Param Control** — Fine-grained Tuning

2 BUSINESS ORCHESTRATION LAYER

Orchestrates task flows, manages asynchronous queues, and injects custom business logic.

- **App Core** — Logic Orchestration
- **Model warm-up** — Cache model
- **Flow Inject** — Dynamic Logic

3 INFERENCE ENGINE LAYER

Executes core neural network computations, model sampling, and efficient pipeline scheduling.

- **ComfyUI** — Graph-Based Core
- **Model** — Resource Mgmt
- **Sampling** — NN Inference

4 HARDWARE ACCELERATION LAYER

Delivers low-level hardware abstraction and optimization for AMD GPUs, ensuring maximum throughput.

- **AMD ROCm** — GPU Compute
- **FP8 Opt** — Precision Boost
- **VRAM Mgmt** — Memory Tuning

Performance Data: A Leap Forward

3×

Inference Speed

Latency reduced from 80s to just 25s, enabling real-time creative workflows.

↓40%

VRAM Usage

Memory footprint optimized from 50G to 30G, fitting larger models into consumer-grade GPUs.

↓90%

OOM Error Rate

From frequent crashes to nearly eliminated, ensuring a stable and uninterrupted experience.

AMD ROCm Optimization Tech Stack

Expandable Memory Segments

Enable expandable memory segments to fundamentally reduce VRAM fragmentation, ensuring efficient memory allocation and sustained performance during prolonged, high-intensity inference workloads.

Full Model Persistence & Warm-Up

Models remain resident in VRAM after initial load, eliminating repetitive loading overhead. Achieves ultra-low latency for subsequent inferences, delivering seamless second-level response speeds.

FP8 Precision Quantization

Halves UNet's VRAM footprint (24GB → 12GB) via FP8 quantization, without sacrificing inference quality. This makes large-scale model deployment feasible on mainstream 48GB consumer-grade GPUs.

Automatic Garbage Collection

Actively forces temporary tensor release after each inference round, proactively cleaning up unused resources. Prevents hidden VRAM leaks and guarantees 24/7 stable, uninterrupted service operation.

Social Value

Lowering the Barrier

Accessible via any browser, it eliminates technical hurdles, making AI-powered creation available to everyone, regardless of skill level or background.

Privacy & Security

Data processing stays local on your device, never touching the cloud. This ensures complete user privacy and data sovereignty from the ground up.

Breaking the Monopoly

Optimized for AMD hardware, it challenges the exclusive ecosystem, delivering premium AI performance and breaking the grip of restrictive hardware monopolies.

Open Source for All

Licensed under the permissive MIT agreement, it encourages community-driven innovation. Free for commercial use and modification, it empowers developers to build, customize, and advance AI tools freely.

Empowering SMBs

Providing professional-grade AI editing capabilities at zero cost levels the playing field. It significantly reduces operational expenses, enabling small and medium businesses to compete with industry giants.

Summary & Outlook

Core Advantages

Complete Workflow
Covers 4 template for a seamless and integrated experience.

Ultimate Performance
3× faster inference speed with 40% reduction in VRAM consumption.

Zero-Friction Ease of Use
One-click installation code, ready to generate content out of the box.

Enterprise-Grade Privacy
100% local execution ensures data never leaves your secure environment.

Future Outlook

Video Frame Stylization
Expand beyond images to support local video editing, enabling real-time style transfer and dynamic visual enhancement directly on video frames.

24GB VRAM Optimization
Launch a dedicated lightweight inference mode for AMD GPUs with 24GB VRAM, breaking hardware barriers for broader accessibility.

Batch Workflow Automation
Introduce a task queue system to handle bulk processing efficiently, significantly boosting productivity for large-scale content generation.

Thank You

GitHub

https://github.com/gao-jie-ai/Track-1-gaojie-AMD-Optimized-Qwen-VL-PartialInpainting#quick-start-guide

It includes complete code, environment configuration guide, and detailed technical documentation.